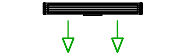
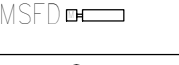
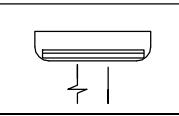
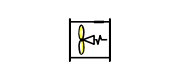
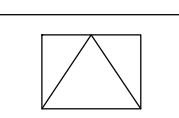
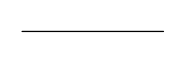
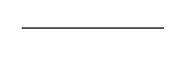
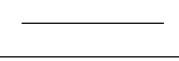
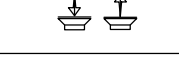
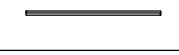
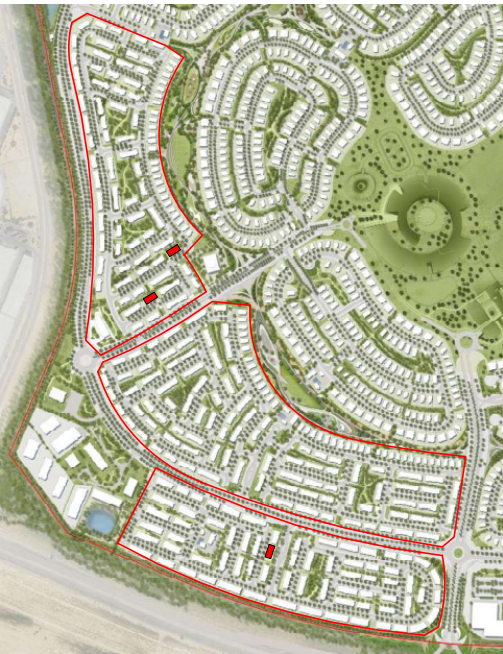


MECHANICAL SYMBOLS AND LEGEND										
SYMBOLS		DESCRIPTION		ABBREV.	DESCRIPTION		ABBREV.	DESCRIPTION		
		RETURN SLOT LINEAR DIFFUSER		SLD	SUPPLY AIR LINEAR SLOT DIFFUSER		IV	ISOLATING VALVE		
		SUPPLY AIR GRILLES AND REGISTERS		RLD	RETURN AIR LINEAR SLOT DIFFUSER			NRV	NON RETURN VALVE	
		SUPPLY SLOT LINEAR DIFFUSER		SCD	SUPPLY CEILING DIFFUSER				BFV	BUTTERFLY VALVE
		RETURN AIR GRILLES AND REGISTERS		RCD	RETURN CEILING DIFFUSER			BFV(M)		BUTTERFLY VALVE (MOTORISED)
		DISC VALVE		ECD	EXHAUST CEILING DIFFUSER		AFV	AUTO-FLOW VALVE		
VCD 		VOLUME CONTROL DAMPER		DV	DISC VALVE			DC	DRAIN COCK	
FD		FIRE DAMPER		MAG	MAKE UP AIR GRILLE		STR		STRAINER	
NRD		NON RETURN DAMPER		SAG	SUPPLY LINEAR BAR GRILLE			PG	PRESSURE GAUGE	
MD 		MOTORISED DAMPER		RAG	RETURN LINEAR BAR GRILLE		CS		COMMISSIONING SET	
MSFD 		MOTORISED SMOKE FIRE DAMPER		DG	DOOR GRILLE (REF. NO.)			TG	TEMPERATURE GAUGE	
		TERMOSTAT		TEMPERATURE CONTROLLER		STL	SANDTRAP LOUVER (LOUVER, REF. NO.)			
		FCU	FAN COIL UNIT		TEMPERATURE SENSOR		CHWF/R	CHILLED WATER FLOW/RETURN		
		WALL MOUNTED TYPE UNIT		SPG	STAIR PRESSURIZATION GRILLE			CHWP	CHILLED WATER PUMP	
		EXAUST FAN		FAL	FRESH AIR LOUVER (LOUVER, REF. NO.)		CWP		CONDENSER WATER PUMP	
		ACCESS PANEL		SAD	SUPPLY AIR DUCT			PHEX	PLATE HEAT EXCHANGER	
		SUPPLY AIR DUCT		RAD	RETURN AIR DUCT		AAV		AUTOMATIC AIR VENT	
		RETURN AIR DUCT		FAD	FRESH AIR DUCT			FCU	FAN COIL UNIT	
		FRESH AIR DUCT		EAD	EXHAUST AIR DUCT		AHU		AIR HANDLING UNIT	
		EXTRACT AIR DUCT		TED	TOILET EXHAUST DUCT			VFD	VARIABLE FREQUENCY DRIVE	
		FRESH/EXTRACT AIR BELL MOUTH		KED	KITCHEN EXHAUST DUCT		PU		PRESSURIZATION UNIT (CHILLED WATER SYSTEM)	
		REFRIGERENT PIPES / CHILLED WATER PIPE		CED	CARPARK EXTRACT DUCT			BMS	BUILDING MANAGEMENT SYSTEM	
				SED	SMOKE EXTRACT DUCT		H/L		HIGH LEVEL	
				VCD	VOLUME CONTROL DAMPER			L/L	LOW LEVEL	
				MD	MOTORISED DAMPER		T/A		TO ABOVE	
				FD	FIRE DAMPER			T/B	TO BELOW	
				AP	ACCESS PANEL		F/A		FROM ABOVE	
				FCU	FAN COIL UNIT			F/B	FROM BELOW	
				AHU	AIR HANDLING UNIT		FJ/FC		FLEXIBLE JOINT / FLEXIBLE CONNECTOR	
				EF	EXHAUST FAN			PS	PRESSURE SWITCH	
				CPEF	CARPARK EXHAUST FAN		OP		ORIFICE PLATE	
				CPSF	CARPARK MAKE-UP AIR FAN			DRV	DOUBLE REGULATING VALVE	
				DPS	DIFFERENTIAL PRESSURE SWITCH		VODRV		VARIABLE ORIFICE DOUBLE REGULATING VALVE	
				FC	FLEXIBLE CONNECTION			FV	FLOAT VALVE	
				AD	ACCESS DOOR (IN DUCT)		FM		FLOW METER	
				CP	CONTROL PANEL			PICV	PRESSURE INDICATING CONTROL VALVE	
				MSFD	MOTORISED FIRE & SMOKE DAMPER		EM		ENERGY METER	
				CAP	CEILING ACCESS PANEL			FODRV	FIXED ORIFICE DOUBLE REGULATING VALVE	
				IU	INDOOR UNIT (SPLIT UNIT)		ATT		ATTENUATOR	
				OU	OUTDOOR UNIT (SPLIT UNIT)			TP	TEST POINT	
				TAD	TRANSFER AIR DUCT		DPCV		DIFFERENTIAL PRESSURE CONTROL VALVE	
				SPF	STAIRCASE PRESSURISATION FAN					
				LPF	LIFT PRESSURIZATION FAN					
				MSF	MAKE-UP AIR SUPPLY FAN					
				SEF	SMOKE EXTRACT FAN					
				SEF	SMOKE EXTRACT FAN					
				SF	SUPPLY FAN					
				KEU	KITCHEN EXTRACT UNIT					
				GEF	GARBAGE ROOM EXTRACT FAN					
				MAD	MAKE-UP AIR SUPPLY DUCT					
				LPD	LIFT PRESSURIZATION DUCT					
				SED	SMOKE EXTRACT DUCT					
				GED	GARBAGE ROOM EXTRACT DUCT					
				SPD	STAIRCASE PRESSURIZATION DUCT					
UNIT ABBREVIATIONS										
ABBREV.	DESCRIPTION		ABBREV.	DESCRIPTION						
L/s	LITER PER SECOND		dB	DECIBEL						
m ³ /s	CUBIC METER PER SECOND		RPM	REVOLUTION PER MINUTE						
m/s	METER PER SECOND		W/kW	WATTS/KILOWATTS						
Pa/kPa	PASCAL/KILOPASCAL		V	VOLTS						
mm	MILLIMETER		PH	PHASE						
m	METER		Hz	HERTZ						
CO	CARBON MONOXIDE		L	LITRE						
°C	DEGREE CELSIUS		PPM	PARTS PER MILLION						
DB	DRY BULB TEMPERATURE		R	RISER						
WB	WET BULB TEMPERATURE		NC	NOISE CRITERIA						
RH	RELATIVE HUMIDITY									
EXTRACT DISC VALVE SCHEDULE										
UNIT NO.	DESCRIPTION		AIR OUTLET SIZE (mm)	CAPACITY RANGE (LPS)						
EDV-01	EXTRACT DISC VALVE		Ø 100	0-12.5 L/s						
EDV-02	EXTRACT DISC VALVE		Ø 200	12.5 -25 L/s						

Key Plan



General Notes :

01. Do not Scale The Drawings. Follow (The MM) Written Dimensions Unless Otherwise Mentioned.
02. Each Drawing Shall be Read in Conjunction With all Other Drawings, Specification & the General Standard Details. In Case of Any Discrepancies Bring it to the Knowledge of Engineer and Same to be Resolved Prior to Execution of Work.
03. Dimensions Shown on These Drawings are Structural Dimensions. Unless Noted Otherwise.
04. All Levels Indicated On These Drawings are Levels as per the the Survey Datum, Same Should be Verified at Site Prior to Execution of Work.
05. All Levels Indicated On These Drawings are Based On Design Level, Which is Equivalent to Datum Level and Same Should be Verified at Site Prior to Execution of Work.



00	10/03/2026	LSG	TENDER SUBMISSION
00	16/01/2026	LSG	DETAILED DESIGN SUBMISSION
00	31/10/2025	LSG	SCHEMATIC DESIGN SUBMISSION
00	15/10/2025	LSG	SCHEMATIC SUBMISSION-50%
00	01/09/2025	LSG	CONCEPT DESIGN SUBMISSION

No.	Date	Initials	Description
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REVISIONS / ISSUE

THIS DRAWING MUST NOT BE REPRODUCED IN WHOLE OR IN PART WITHOUT THE PRIOR APPROVAL OF ARIF & BINTOAK

Status
TENDER SUBMISSION

Client

EMAAR

Lead Consultant



Project Title:
Proposed Residential Townhouses
and Standalone Villas
TH-1, TH-2, and TH-3 at Grand Polo
Club & Resort, Dubai, UAE

Discipline :
MECHANICAL

Drawing title:

HVAC GENERAL NOTES AND LEGEND

Architectural Engineer	Electrical Engineer	Structural Engineer
MUSTAFA ABID	MARWAN MOHD.	RIHAM RAMADAN
Registration # 5847	Registration # 80102	Registration # 105092
Scale(s): NTS	Date: 10/03/2026	Project no: 1857
Drawn by: LSG	Checked: MAV	Approved: MRM

Drawing Number :
GPCR-TH01-AB-DRG-ZZZ-MEC-HV-1002-00